

# How Much Does the Activation Function Matter? A 2,000-Run Variance Study of Gated Recurrent Networks for Sentiment Classification

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**Abstract**—Activation-function comparisons in gated recurrent networks are almost always reported from a single training run. It has therefore never been established whether the differences they report are larger than the noise of retraining the same model. This paper settles that question at scale. We train all 100 ordered pairs of ten activation functions in the dense layers of four gated recurrent architectures on IMDb, and repeat every one of the 400 configurations under five random seeds, for 2,000 models in total with every other setting held fixed. A variance decomposition over these runs shows that the random seed accounts for 58.7% of all variation in test accuracy and the architecture for 26.7%, while the activation pair explains only 3.7%. The consequence is direct: single-seed activation rankings do not reproduce. Across the four architectures the Spearman correlation between the single-seed ranking of the 100 pairs and their five-seed means is  $-0.174$ ,  $-0.059$ ,  $-0.041$  and  $+0.055$ , indistinguishable from zero in every case, and the ten best configurations overlap within their confidence intervals. What survives is architectural: GRU outperforms LSTM by 0.79 points ( $p < 10^{-100}$ , Cohen’s  $d = 1.19$ ), supplies all fifty highest-scoring configurations, and is the more reliable choice, with roughly 40% lower seed-to-seed standard deviation. One activation effect also survives: smooth functions reach 73.7% validation accuracy after one epoch against 61.1% for the ReLU family. We give the seed budget needed to resolve each remaining effect, and argue that grid searches of this kind should report variance as standard.

**Index Terms**—Sentiment Analysis, IMDb Dataset, Deep Learning, LSTM, GRU, Activation Functions, Reproducibility, Seed Variance, Experimental Methodology.

## I. INTRODUCTION

Sentiment analysis is the task of deciding whether a piece of text is positive or negative. It is one of the most common problems in natural language processing. The IMDb Large Movie Review Dataset, with 50,000 balanced reviews, is the standard benchmark for the binary version of the task [16]. LSTM and GRU networks are the usual choice for this problem because their gating mechanisms let them track context across long sequences [1]–[3].

One part of these models gets little attention: the activation function in the dense layers that sit after the recurrent layers.

The gates inside the recurrent cell are fixed to sigmoid and tanh, but the dense layers that turn the recurrent output into a final prediction can use any activation function. Ahmed et al. [17] tested three classical combinations in a bidirectional LSTM, reached 88.40% accuracy, and asked for follow-up work on modern activation functions and other architectures.

Answering that call runs into a problem. Every comparison we are aware of, including all of [17], [18], [21]–[23], trains each configuration once. Training is not deterministic. The weight initialisation, the dropout masks and the order of the training batches all change the final accuracy, even when the data and every setting are held identical. If that change is as large as the gaps between activation functions, then a ranking built from one run per function is measuring the seed rather than the function. No study in this area reports how large that change is, so there is no way to tell which is being measured.

We therefore designed this study around replication. We train all 100 ordered pairs of ten activation functions across four architectures (LSTM, GRU, and the fully bidirectional version of each), then repeat every one of those 400 configurations under five random seeds, holding the data split fixed so that only initialisation, dropout and batch order vary. The result is **2,000 trained models**. To our knowledge this is the largest controlled activation study in gated recurrent networks, and the first in this setting to report seed variance. Five runs of every configuration let us measure how much accuracy moves when only the seed changes, and compare that against the gaps between activation functions:

- **RQ1:** How is accuracy variance divided between architecture, activation choice and seed?
- **RQ2:** Does the architecture exceed seed noise?
- **RQ3:** Does the activation pair exceed seed noise?
- **RQ4:** Does a single-seed ranking reproduce?
- **RQ5:** Which activation effects survive?
- **RQ6:** How many seeds would the rest require?

## II. LITERATURE REVIEW

Hochreiter and Schmidhuber [1] introduced the LSTM to solve the vanishing gradient problem in recurrent networks,

using sigmoid and tanh as internal gate activations. Cho et al. [2] proposed the GRU, which reduces the gating structure to two gates and merges the cell and hidden states, and Chung et al. [3] showed that GRU matches LSTM on sequence tasks with fewer parameters. Schuster and Paliwal [4] added bidirectionality, letting the model use both past and future context. None of these works asked whether the activation functions in the layers around the recurrent cells were optimal.

The activation function literature evolved separately. Nair and Hinton [5] introduced ReLU, trading saturation for the risk of killing neurons on negative inputs; Maas et al. [6] and Clevert et al. [8] addressed this with LeakyReLU and ELU, and Klambauer et al. [9] introduced SELU with a self-normalising property. More recently GELU [10] became the transformer default, Swish [11] was found by automated search, and Mish [12] outperformed both on vision benchmarks; surveys [14], [15] catalogue the growing range. All were developed and tested in feedforward, convolutional or transformer settings, leaving their behaviour inside gated recurrent dense layers untested.

Within sentiment analysis the IMDB dataset [16] is the standard binary benchmark; Tang et al. [20] validated GRU for document-level sentiment and Socher et al. [19] contributed the SST treebank. Several studies addressed the activation question partially. Eger et al. [21] compared 21 activations across NLP tasks but used basic architectures and predated Mish. Farzad et al. [22] swept 23 functions at the LSTM gate level on IMDB with no GRU or bidirectional variants. Essai Ali et al. [23] engineered 26 activations for LSTM, but outside sentiment and without GELU, Swish or Mish. Zhu and Tan [18] tested a custom activation on LSTM and GRU for IMDB against only ReLU and SELU. Ahmed et al. [17] tested three classical pairings in a BiLSTM on IMDB, reached 88.40%, and called for broader architectures and modern functions.

Four gaps remain open, and the fourth subsumes the others. No study has swept activation pairings exhaustively rather than testing a few hand-picked combinations. No study has compared across LSTM, GRU, and their fully bidirectional variants under identical conditions. No study has benchmarked GELU, SiLU, or Mish inside gated recurrent dense layers for sentiment classification.

Most importantly, *none of the works above repeats any configuration*. Eger et al. [21], Farzad et al. [22], Essai Ali et al. [23], Zhu and Tan [18] and Ahmed et al. [17] each report a single run per setting, and none reports a standard deviation, a confidence interval, or a seed. Their rankings are therefore single draws from a stochastic process whose spread is unknown. Reproducibility studies in other areas of deep learning have found such rankings to be fragile, but the question has not been put to activation functions in gated recurrent networks. This paper addresses all four gaps, and treats the fourth as its primary contribution.

### III. METHODOLOGY

#### A. Dataset and Preprocessing

The IMDB Large Movie Review Dataset [16] (50,000 reviews, balanced binary labels) was split 80/20 train-test

with `random_state=42`, and 10% of training data was withheld as a validation set. Reviews were cleaned by removing HTML tags and non-alphabetic characters, then lowercased. A vocabulary of the 5,000 most frequent training-set tokens was used; all sequences were truncated or zero-padded to 200 tokens.

#### B. Network Architecture

All four experiments share the same model structure. The only thing that changes within each experiment is the choice of  $\text{Act}_1$  and  $\text{Act}_2$  in the two dense layers. The shared architecture is: an embedding layer (vocabulary 5,000, dimension 128), a bidirectional recurrent Layer 1 (64 units per direction), a recurrent Layer 2 (32 units per direction), Dropout(0.4), Dense Layer 1 (64 units) with  $\text{Act}_1$ , Dense Layer 2 (32 units) with  $\text{Act}_2$ , and a single output unit trained with BCEWithLogitsLoss. The four experiments differ only in the recurrent cell type (LSTM, GRU) and whether Layer 2 is bidirectional (Table I).

TABLE I  
EXPERIMENTAL ARCHITECTURE SUMMARY

| Exp. | Layer 1    | Layer 2    | Params                |
|------|------------|------------|-----------------------|
| 1    | BiLSTM(64) | LSTM(32)   | $\approx 764\text{K}$ |
| 2    | BiGRU(64)  | GRU(32)    | $\approx 734\text{K}$ |
| 3    | BiLSTM(64) | BiLSTM(32) | $\approx 787\text{K}$ |
| 4    | BiGRU(64)  | BiGRU(32)  | $\approx 752\text{K}$ |

#### C. Activation Functions Under Investigation

Ten activation functions were evaluated (Table II). The classical family is ReLU [5], LeakyReLU [6], PReLU [7], ELU [8], Tanh and SELU [9]. The modern smooth family is GELU [10], SiLU (Swish) [11], Mish [12] and Hardswish [13]. The latter four had not previously been benchmarked in gated recurrent dense layers for sentiment classification.

TABLE II  
ACTIVATION FUNCTIONS: FAMILIES AND KEY PROPERTIES

| Function  | Type      | Formula / Key Property                            |
|-----------|-----------|---------------------------------------------------|
| ReLU      | Classical | $\max(0, x)$ ; dead neurons for $x < 0$           |
| ELU       | Classical | $\alpha(e^x - 1)$ for $x \leq 0$ ; mean-centering |
| LeakyReLU | Classical | $0.01x$ for $x < 0$ ; low dead-neuron risk        |
| Tanh      | Classical | Bounded; saturates for $ x  \gg 1$                |
| GELU      | Modern    | $x \cdot \Phi(x)$ ; probabilistic gating          |
| SiLU      | Modern    | $x \cdot \sigma(x)$ ; self-gating                 |
| Mish      | Modern    | $x \cdot \tanh(\text{softplus}(x))$ ; smooth      |
| SELU      | Modern    | Self-normalising; implicit batch norm             |
| PReLU     | Modern    | Learnable slope $a$ ; input-dependent             |
| Hardswish | Modern    | Piecewise; dead zone for $x < -3$                 |

#### D. Training Configuration and Evaluation

Every setting in Table III was held the same across all runs, so the activation pair is the only thing that changes between

models. The architecture parameters (vocabulary, embedding, layer sizes, dropout) replicate Ahmed et al. [17] exactly. The training parameters (optimizer, learning rate, batch size, scheduler, clipping) were chosen by us because Ahmed et al. did not report them. After training, accuracy, precision, recall, and F1-score were measured on the held-out 10,000-review test set.

#### E. Replication Protocol

Each of the  $10 \times 10 \times 4 = 400$  configurations was trained under five seeds,  $\{10, 20, 30, 40, 50\}$ , giving 2,000 models. The seed is applied immediately before the model and its data loaders are constructed, setting `random`, `numpy`, `torch` and `torch.cuda` with `cudnn.deterministic=True`, so a given (pair, seed) is exactly reproducible.

The split is deliberately *not* a function of the seed: it stays pinned at `random_state=42`. The variance we report is therefore the variance of retraining the same model on the same data. That is the quantity a single-run comparison assumes to be negligible. Split variance would only inflate it, so these estimates are conservative.

Runs were executed five at a time, one worker per seed. Each worker is a separate process with its own RNG and deterministic kernel selection, so concurrency cannot perturb a result; we confirmed this by re-running sampled configurations serially and recovering identical metrics.

TABLE III  
SHARED HYPERPARAMETER CONFIGURATION

| Hyperparameter                | Value                              | Source          |
|-------------------------------|------------------------------------|-----------------|
| Vocab / Seq. length / Embed.  | 5,000 / 200 / 128                  | [17]            |
| RNN Layer 1 / Layer 2 units   | 64 / 32 per direction              | [17]            |
| Dense Layer 1 / Layer 2 units | 64 / 32                            | [17]            |
| Dropout rate                  | 0.4                                | [17]            |
| Batch size / Max epochs       | 64 / 10                            | Ours            |
| Early stopping patience       | 3 epochs (val. loss)               | Ours            |
| Optimiser                     | Adam, $\text{lr} = 10^{-3}$        | Ours            |
| LR scheduler                  | ReduceLROnPlateau ( $\times 0.5$ ) | Ours            |
| Loss / Gradient clipping      | BCEWithLogitsLoss / $L_2 \leq 1.0$ | Ours            |
| Weight initialisation         | Kaiming uniform (Linear)           | PyTorch default |
| Data split seed               | 42, fixed for all runs             | Ours            |
| Training seeds                | 10, 20, 30, 40, 50                 | Ours            |

#### F. On Reproducing the Baseline of Ahmed et al.

Ahmed et al. [17] describe their architecture in full and we copy it exactly, but they do not report how they trained it: optimizer, learning rate, batch size, epochs, early stopping, seed and data split are all missing. Exact reproduction is therefore impossible, and we adopt the fully documented settings of Table III instead.

Replication sharpens what can be said about their 88.40%. Our best configuration averages 87.81% with a 95% upper bound of 88.01%, and no run among the 2,000 reached 88.40% (maximum 88.39%), so the benchmark sits outside the interval of our strongest setting. Since they report a single run without a seed or standard deviation, and we measure the per-cell standard deviation of this model family at 0.38 points, their

figure is best read as one draw of unknown uncertainty. We treat it as a reference point, not a target.

## IV. RESULTS

### A. Where the Variance Comes From

Test accuracy differs from run to run for three reasons: the architecture, the activation pair, and the random seed. Table IV splits the total variance across the 2,000 runs between them.

The seed accounts for **58.7%**. This is variance between runs that differ in nothing but the seed, so no choice of architecture or activation removes it. The architecture accounts for 26.7%, almost all of it (26.3%) from the choice of LSTM or GRU cell; making the second recurrent layer bidirectional accounts for 0.19%. The activation pair accounts for **3.7%**.

TABLE IV  
VARIANCE DECOMPOSITION OF TEST ACCURACY (2,000 RUNS)

| Source of variation                                   | % of total variance |
|-------------------------------------------------------|---------------------|
| Architecture (4 levels)                               | 26.69               |
| cell type (GRU vs. LSTM)                              | 26.32               |
| layer-2 bidirectionality                              | 0.19                |
| Act <sub>1</sub> (10 levels)                          | 0.53                |
| Act <sub>2</sub> (10 levels)                          | 0.55                |
| Act <sub>1</sub> × Act <sub>2</sub> pair (100 levels) | 3.74                |
| <b>Seed (residual)</b>                                | <b>58.67</b>        |

The activation effect is about sixteen times smaller than the seed effect. A study that trains each configuration once measures the two together and cannot separate them. Section V-C gives the number of seeds each effect needs.

### B. Classification Performance

Fig. 1 plots the mean accuracy of each of the 400 configurations, grouped by architecture. Every architecture mean comes from 500 runs, which gives a 95% confidence interval of  $\pm 0.03$  to  $\pm 0.10$  points.

The two GRU architectures both average 87.48% and cannot be told apart. Both are above the two LSTM architectures, at 86.62% and 86.75%. Within each architecture the 100 activation pairs form one narrow cloud, and the gap between the GRU and LSTM clouds is wider than any cloud. Changing the cell type moves accuracy more than changing the activation pair does.

Pooling by cell type over 1,000 runs each, GRU reaches  $87.48 \pm 0.02\%$  and LSTM  $86.69 \pm 0.05\%$ . The gap of **0.79** points is large relative to its uncertainty (Welch  $t = 26.7$ ,  $p \approx 2 \times 10^{-125}$ , Cohen's  $d = 1.19$ ). No configuration's mean reaches the 88.40% benchmark of Ahmed et al. [17]; the best cell mean is 87.81% with a 95% upper bound of 88.01%, and none of the 2,000 individual runs attained 88.40% (maximum 88.39%). We therefore treat that figure as unmatched under our documented training settings rather than as a target we reached.

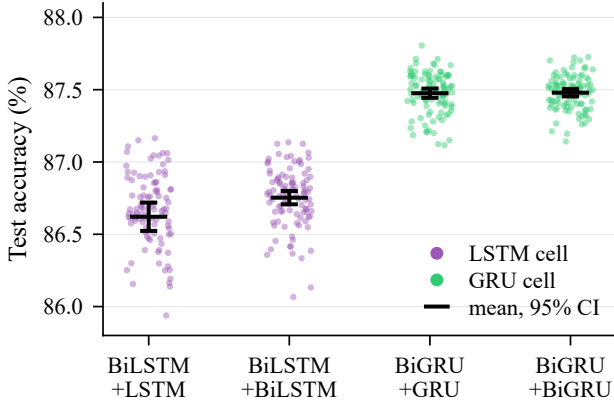


Fig. 1. The 400 replicated cell means, grouped by architecture. Each point is one activation pair averaged over five seeds; black markers give the architecture mean with its 95% confidence interval. The  $y$ -axis is truncated at 85.8%, which clips one outlier cell (Hardswish + LeakyReLU in BiLSTM+LSTM, 82.06%,  $\sigma = 9.27$ ).

### C. The Ten Best Configurations

Table V lists the ten highest-scoring configurations of the 400, each with its standard deviation and 95% confidence interval. All ten use a GRU cell, as do the top fifty.

The ten means lie between 87.68% and 87.81%, a spread of **0.12** points. Their confidence intervals are wider than that, up to  $\pm 0.50$  points. Every interval overlaps every other, and one range of values is consistent with all ten at once. *Nothing in the data separates these ten configurations*, so the order of the rows in Table V is not a ranking.

TABLE V  
TEN HIGHEST-ACCURACY CONFIGURATIONS OF 400, WITH SEED VARIABILITY

| Act <sub>1</sub> + Act <sub>2</sub> | Experiment  | Acc.%<br>mean $\pm \sigma$ | 95% CI<br>$\pm$ | F1%<br>mean |
|-------------------------------------|-------------|----------------------------|-----------------|-------------|
| ReLU + LeakyReLU                    | BiGRU+GRU   | 87.81 $\pm$ 0.17           | 0.21            | 87.80       |
| SiLU + Mish                         | BiGRU+BiGRU | 87.73 $\pm$ 0.16           | 0.20            | 87.65       |
| Tanh + LeakyReLU                    | BiGRU+BiGRU | 87.72 $\pm$ 0.40           | 0.50            | 87.70       |
| Tanh + GELU                         | BiGRU+GRU   | 87.71 $\pm$ 0.29           | 0.36            | 87.75       |
| PReLU + ReLU                        | BiGRU+GRU   | 87.71 $\pm$ 0.17           | 0.22            | 87.67       |
| GELU + GELU                         | BiGRU+BiGRU | 87.70 $\pm$ 0.18           | 0.22            | 87.58       |
| SiLU + Hardswish                    | BiGRU+BiGRU | 87.70 $\pm$ 0.10           | 0.12            | 87.69       |
| GELU + Hardswish                    | BiGRU+BiGRU | 87.69 $\pm$ 0.18           | 0.23            | 87.65       |
| SiLU + GELU                         | BiGRU+GRU   | 87.68 $\pm$ 0.18           | 0.22            | 87.63       |
| PReLU + GELU                        | BiGRU+GRU   | 87.68 $\pm$ 0.28           | 0.34            | 87.71       |

All ten are GRU-based. The ten means span 0.12 points, less than the widest half-interval ( $\pm 0.50$ ); every pair of intervals overlaps.

### D. Reproducibility of the Single-Seed Ranking

We ranked the 100 activation pairs in each architecture twice: once by the accuracy of the original single run, and once by the five-seed mean. Table VI compares the two rankings.

The Spearman correlation between the two rankings is  $-0.174$ ,  $-0.059$ ,  $-0.041$  and  $+0.055$  in the four architectures. A correlation of zero means the first ranking says nothing about the second, and all four values are close to zero. A pair moves 32 to 37 places out of 100 on average between the two

TABLE VI  
AGREEMENT BETWEEN THE SINGLE-SEED RANKING AND THE 5-SEED MEANS

| Experiment    | $\rho$   | Mean rank $\Delta$ | Top-10 overlap | Mean $ \Delta\text{acc} $ |
|---------------|----------|--------------------|----------------|---------------------------|
| BiLSTM+LSTM   | $-0.174$ | 36.9               | 0/10           | 0.53                      |
| BiGRU+GRU     | $-0.059$ | 34.5               | 1/10           | 0.34                      |
| BiLSTM+BiLSTM | $-0.041$ | 34.4               | 1/10           | 0.45                      |
| BiGRU+BiGRU   | $+0.055$ | 32.1               | 0/10           | 0.36                      |

$\rho$  is the Spearman correlation between the 100 single-seed cell accuracies and the corresponding 5-seed means.  $\rho \approx 0$  in all four.

lists. Of the ten pairs that led each single-seed sweep, at most one is still in the top ten after replication.

The cause is selection bias. A pair reaches the top of a single-seed sweep partly because it is good and partly because its one run was lucky. LeakyReLU + Tanh led the BiGRU+GRU sweep at 88.05%. Its five runs are 86.70, 87.13, 87.45, 87.46 and 88.20, so the published figure was its best draw; its mean is  $87.39 \pm 0.55$ , which ranks 77th of 100. The pair that leads on the mean, ReLU + LeakyReLU, scores  $87.81 \pm 0.17$ . The single-seed winners of the other three architectures fall to ranks 98, 91 and 85. Averaged over the ten leaders of the original sweep, replication costs 0.48 points.

### E. Marginal Effects per Activation

Averaging each function over all ten of its partners and all five seeds gives 50 runs per estimate, which removes most of the noise. A small signal appears at that level: the ten averages within one architecture span at most 0.67 points, and within BiGRU+GRU only 0.18.

The orderings still do not agree between architectures. Comparing the four architectures pairwise gives six rank correlations, from  $-0.164$  to  $+0.576$ , none significant at the 5% level ( $p \geq 0.082$ ). A single-seed reading of this grid suggested that a function good in one architecture was bad in another, with ELU as the example: best Act<sub>1</sub> in BiGRU+GRU and last in BiLSTM+LSTM. Replicated, ELU ranks 9th of ten in BiGRU+GRU and 3rd in BiLSTM+LSTM, so both halves of that claim reverse. The orderings are unrelated rather than inverted. Per-function averages for all four architectures are given in the supplementary material.

### F. Convergence Speed

The main sweep records only final accuracy. To measure learning speed we re-ran the ten homogeneous pairs (X+X) on BiGRU+GRU with per-epoch logging under five seeds (Table VII, Fig. 2).

This is the one activation effect in the study that survives replication. After a single epoch the smooth family (Mish, GELU, SiLU, Hardswish) averages **73.70%** validation accuracy against **61.05%** for the ReLU family (ReLU, LeakyReLU, SELU). The 12.65-point gap is 2.9 times the median per-function seed standard deviation of 4.39, and the ordering is preserved against the single-seed measurement (Spearman  $\rho = 0.891$ ,  $p = 0.0005$ ). The advantage is in speed only: by

TABLE VII  
EPOCH-1 CONVERGENCE ON BiGRU+GRU, FIVE SEEDS

| Activation | Val. loss (E1) | Val. acc. (E1)% | $\sigma$ |
|------------|----------------|-----------------|----------|
| Hardswish  | 0.523          | <b>74.55</b>    | 4.05     |
| GELU       | 0.541          | 74.21           | 3.32     |
| Mish       | 0.544          | 74.03           | 5.97     |
| SiLU       | 0.558          | 72.00           | 3.95     |
| PReLU      | 0.625          | 66.87           | 2.22     |
| Tanh       | 0.651          | 62.76           | 7.52     |
| LeakyReLU  | 0.655          | 62.40           | 4.73     |
| ELU        | 0.644          | 62.20           | 7.29     |
| ReLU       | 0.652          | 61.51           | 7.50     |
| SELU       | 0.674          | 59.24           | 3.73     |

Smooth family (Mish, GELU, SiLU, Hardswish) averages 73.70%; the ReLU family (ReLU, LeakyReLU, SELU) averages 61.05%.

epoch three all functions reach comparable validation loss, and the final test accuracies are within noise of each other.

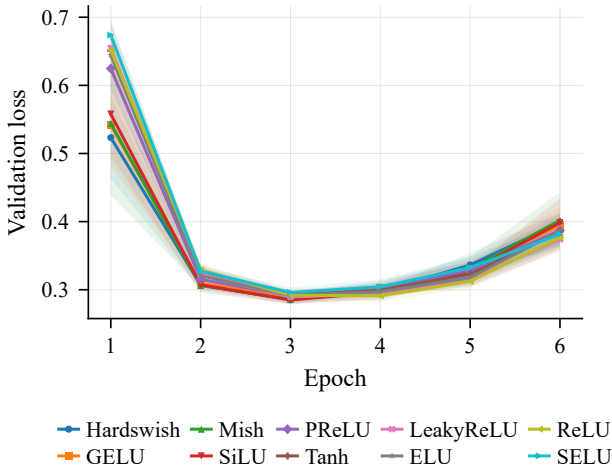


Fig. 2. Per-epoch validation loss on BiGRU+GRU for the ten homogeneous pairs. Curves are means over five seeds; shaded bands are  $\pm 1$  standard deviation. The separation visible at epoch 1 has closed by epoch 3.

### G. Reliability and Failure Modes

Accuracy is not the only axis on which the architectures differ. The median per-cell standard deviation is 0.30 for the GRU family against 0.50 for the LSTM family ( $p \approx 2 \times 10^{-7}$ ): GRU is both more accurate and about 40% more stable run-to-run. Instability is also concentrated. Of the eight cells whose standard deviation exceeds 1.0 point, *all eight* are LSTM-family. The extreme case, Hardswish + LeakyReLU in BiLSTM+LSTM, averages 82.06% with  $\sigma = 9.27$  and a range of 65.50–86.85%: it trains normally under four seeds and collapses under the fifth. Apart from that collapse, no run of the 2,000 fell below 60% accuracy.

Two claims from the single-seed reading do not survive. Per-cell precision–recall imbalance is one: the widest gap between mean precision and mean recall over any cell is 6.89 points while individual runs reach 24.17, so large gaps belong to runs, not to configurations. A dense-layer ReLU penalty is the

other; in BiLSTM+LSTM, ReLU is the strongest Act<sub>1</sub> choice (86.83% versus 86.60%,  $p = 0.006$ ).

The campaign cost about six GPU-hours on one NVIDIA RTX 5080 at 13.4 s per run. GRU models carry 3.8% fewer parameters than their LSTM counterparts (734K versus 764K), making BiGRU+GRU the cheapest, joint most accurate and most stable of the four.

## V. DISCUSSION

### A. RQ1–RQ3: What Actually Moves Accuracy

We have not found a prior estimate of the variance split for gated recurrent networks. It sets a limit on what a single-run study can show: an effect worth 3.7% of the variance cannot be read off a design whose noise is worth 58.7%.

The cell-type effect is large next to the noise, at  $d = 1.19$ , and four seeds per cell are enough to detect it. It is also the one conclusion of the original single-seed design that replication leaves intact: the five-seed architecture means differ from the single-seed means by at most 0.18 points. Layer-2 bidirectionality is the opposite case. It accounts for 0.19% of the variance, and adds 0.13 points in LSTM ( $p = 0.017$ ) and 0.00 points in GRU ( $p = 0.858$ ). In the GRU case, which is the faster and more accurate family, the extra parameters add nothing.

The activation pair separates from the noise only when many runs are averaged. Within BiGRU+GRU the 100 cell means span 0.69 points, while a single cell has a standard deviation of 0.33 across its seeds. Averages over ten partners therefore separate; individual pairs do not. The gap between the LSTM and GRU families, 0.79 points, is larger than the whole 0.69-point span of activation choices inside GRU.

### B. RQ4: Single-Seed Rankings Do Not Reproduce

In all four architectures the single-seed and five-seed rankings of the 100 activation pairs are unrelated (Table VI).

The models themselves are stable. Almost every run lands in a narrow band, and only one configuration in 400 failed to train reliably. The problem is the size of the gaps being ranked. Two pairs adjacent in the ranking differ by about 0.005 points, while a single pair varies by 0.29 to 0.53 points across its own five seeds. The noise is roughly eighty times the gap, so one run per pair sorts seeds rather than functions.

### C. RQ5–RQ6: What Survives, and at What Cost

One activation effect survives cleanly: early convergence. Smooth functions reach 73.70% validation accuracy after one epoch against 61.05% for the ReLU family, 2.9 times the seed standard deviation with the ordering preserved ( $p = 0.891$ ). Where the budget is measured in epochs, that is a reason to prefer Mish, GELU, SiLU or Hardswish. A second effect survives in weaker form: the GRU family is more stable, and every high-variance cell in the grid is LSTM-based. Reliability is rarely reported when architectures are compared; here it points the same way as accuracy. What does not survive is the identity of the best pair, the transfer or inversion of rankings

across architectures, per-cell precision–recall behaviour, and any penalty for dense-layer ReLU.

The number of seeds needed to resolve each effect follows from its size. Taking the median per-cell standard deviation of 0.38 points, at 80% power and  $\alpha = 0.05$ , detecting the 0.79-point architecture effect needs about 4 seeds per cell, which is why five were enough here. Separating the top-ranked cell from the tenth, a gap of 0.12 points, needs about 157 seeds per cell. Separating cells adjacent in the top ten needs several thousand. At the measured 13.4 s per run, the 157-seed case alone is about 230 GPU-hours for one architecture.

This is the answer to which activation pair is best. It is not that the winner escaped us: the gaps are small enough that identifying one would take a budget far beyond what studies in this area spend, and naming a winner from a single run measures the seed.

#### D. Limitations

The study covers one dataset and one task family, so the variance split should be re-estimated before being assumed elsewhere. The data split is held fixed by design, so our estimates exclude split variance and are lower bounds. Five seeds suffice for the architectural conclusions, but leave per-cell intervals wide relative to the differences between cells, so individual cell means should be read with their intervals attached.

### VI. CONCLUSION

We trained 2,000 models: all 100 ordered pairs of ten activation functions, in four gated recurrent architectures, each replicated under five seeds. The question was not which activation function wins, but whether that question is answerable as it is usually posed.

Five findings follow. *First*, the random seed explains 58.7% of the variance in test accuracy, the architecture 26.7%, and the activation pair 3.7%: seed variation dominates every controlled factor combined. *Second*, single-seed rankings of activation pairs do not reproduce. Across four independently swept architectures the correlation between the single-seed ordering of 100 cells and their replicated means is  $-0.174$ ,  $-0.059$ ,  $-0.041$  and  $+0.055$ , with average rank movement above 32 places. The ten best configurations of 400 span 0.12 points against intervals up to  $\pm 0.50$ , making them statistically indistinguishable. *Third*, the architectural conclusion is large and robust: GRU exceeds LSTM by 0.79 points ( $p \approx 2 \times 10^{-125}$ ,  $d = 1.19$ ) and supplies all fifty highest-scoring configurations, while layer-2 bidirectionality contributes 0.19% of variance and nothing measurable for GRU. *Fourth*, GRU is also the more reliable family, with about 40% lower seed-to-seed standard deviation, and all eight high-variance cells in the grid are LSTM-based. *Fifth*, one activation effect survives replication: smooth functions reach 73.70% validation accuracy after one epoch against 61.05% for the ReLU family, 2.9 times the seed standard deviation. The identity of the best pair, the cross-architecture inversion of rankings, and the dead-neuron penalty for dense-layer ReLU do not.

For practitioners: choose GRU, prefer a smooth activation if the budget is measured in epochs, and do not tune the pair further. For the field: report variance.

Future work should test whether this variance split holds on other sentiment corpora. SST-2 and Amazon Reviews are the natural next targets, and would establish whether the architectural advantage generalises across domains. The same question applies to attention-based models, where activation choice in feed-forward blocks follows the same single-run reporting convention.

### REFERENCES

- [1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [2] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. EMNLP*, 2014, pp. 1724–1734.
- [3] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, “Empirical evaluation of gated recurrent neural networks on sequence modeling,” *arXiv preprint arXiv:1412.3555*, 2014.
- [4] M. Schuster and K. K. Paliwal, “Bidirectional recurrent neural networks,” *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.
- [5] V. Nair and G. E. Hinton, “Rectified linear units improve restricted Boltzmann machines,” in *Proc. ICML*, 2010, pp. 807–814.
- [6] A. L. Maas, A. Y. Hannun, and A. Y. Ng, “Rectifier nonlinearities improve neural network acoustic models,” in *Proc. ICML Workshop on Deep Learning for Audio, Speech and Language Processing*, 2013.
- [7] K. He, X. Zhang, S. Ren, and J. Sun, “Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification,” in *Proc. ICCV*, 2015, pp. 1026–1034.
- [8] D. Clevert, T. Unterthiner, and S. Hochreiter, “Fast and accurate deep network learning by exponential linear units (ELUs),” *arXiv preprint arXiv:1511.07289*, 2015.
- [9] G. Klambauer, T. Unterthiner, A. Mayr, and S. Hochreiter, “Self-normalizing neural networks,” in *Proc. NeurIPS*, 2017, pp. 971–980.
- [10] D. Hendrycks and K. Gimpel, “Gaussian error linear units (GELUs),” *arXiv preprint arXiv:1606.08415*, 2016.
- [11] P. Ramachandran, B. Zoph, and Q. V. Le, “Searching for activation functions,” *arXiv preprint arXiv:1710.05941*, 2017.
- [12] D. Misra, “Mish: A self regularized non-monotonic activation function,” in *Proc. BMVC*, 2020.
- [13] A. Howard *et al.*, “Searching for MobileNetV3,” in *Proc. ICCV*, 2019, pp. 1314–1324.
- [14] A. Apicella, F. Donnarumma, F. Isgrò, and R. Prevete, “A survey on modern trainable activation functions,” *Neural Networks*, vol. 138, pp. 14–32, 2021.
- [15] C. Nwankpa, W. Ijomah, A. Gachagan, and S. Marshall, “Activation functions: Comparison of trends in practice and research for deep learning,” *arXiv preprint arXiv:1811.03378*, 2018.
- [16] A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, “Learning word vectors for sentiment analysis,” in *Proc. ACL*, 2011, pp. 142–150.
- [17] S. A. Ahmed *et al.*, “Comparative analysis of activation functions in LSTM models for sentiment classification,” *IJARCCCE*, vol. 14, no. 4, Apr. 2025.
- [18] Y. Zhu and R. Tan, “A novel activation function based recurrent neural networks and their applications on sentiment classification and log anomaly detection,” *Frontiers in Neurorobotics*, vol. 16, 2022.
- [19] R. Socher, A. Perelygin, J. Wu, J. Chuang, C. D. Manning, A. Ng, and C. Potts, “Recursive deep models for semantic compositionality over a sentiment treebank,” in *Proc. EMNLP*, 2013, pp. 1631–1642.
- [20] D. Tang, B. Qin, and T. Liu, “Document modeling with gated recurrent neural network for sentiment classification,” in *Proc. EMNLP*, 2015, pp. 1422–1432.
- [21] S. Eger, P. Youssef, and I. Gurevych, “Is it time to swish? Comparing deep learning activation functions across NLP tasks,” in *Proc. EMNLP*, 2018, pp. 4423–4433.
- [22] A. Farzad, H. Mashayekhi, and H. Hassanpour, “A comparative performance analysis of different activation functions in LSTM networks for classification,” *Neural Computing and Applications*, vol. 31, pp. 2507–2521, 2019.
- [23] M. H. Essai Ali, A. I. Taloba, M. F. Farhan, and R. A. Abozeid, “Developing novel activation functions based deep learning LSTM for classification,” *IEEE Access*, vol. 10, pp. 96578–96591, 2022.